Razieh Ghaedi

Incoming PhD Candidate: Cardiff Business School

07960177520 rozaghaedi90@gmail.com LinkedIn GitHub Google Scholar Sheffield, UK

Professional Summary

Motivated MSc Computer Scientist (Distinction) with a strong research background in deep learning, multimodal learning, and human-centered AI. Experienced in designing and evaluating data-driven models for cross-domain facial expression recognition, affective computing, and socially responsible AI applications. Demonstrated research output through peer-reviewed publications (Knowledge-Based Systems, ACML 2025, IEEE Access) and interdisciplinary collaboration at the University of Sheffield. Research interests encompass machine learning, deep learning, and AI applications, with particular focus on interpretable, robust, and socially responsible intelligent systems.

Education

PhD in Management (AI for Health Equity) — Oct 2026 – Sep 2029

Cardiff Business School, Cardiff University — *Cardiff, UK*

- Recipient of CARBS (Cardiff Business School) PhD Studentship
- Project: *"Reducing Health Screening Inequalities in Wales: AI-Driven Forecasting and Resource Optimisation for Underserved Communities"*
- **Supervisors:** Prof. Bahman Rostami-Tabar, Prof. Paul Harper

MSc in Computer Science — Sep 2023 – Oct 2024

Manchester Metropolitan University — *Manchester, UK*

- Graduated with a Distinction
- Dissertation: Deep Learning-Based Cross-Domain Facial Expression Recognition

BSc in Information Technology Engineering — Sep 2009 – Feb 2014

- **Key Modules:** Data Structures, Algorithms Design, Advanced Programming, Operating Systems, Database Design, Artificial Intelligence, Digital Electronics, Software Engineering

Publications

- **Ghaedi, R.**, BabaAhmadi, A., Alam, N., Fan, X. (2025). ***Graph-Attention Network with Adversarial Domain Alignment for Robust Cross-Domain Facial Expression Recognition***. Presented at **ACML 2025**. [**Paper**] | [**Poster**]
- Zarina, O., Saleh Pour, M. J., **Ghaedi, R.**, Shafei Khah, M. (2023). ***Markov-Based Reliability Assessment for Distribution Systems Considering Failure Rates***. **IEEE Access**, 11, 10018–10031. DOI:10.1109/ACCESS.2023.10032136
- Shirdel, M., **Ghaedi, R.**, Fan, X. (2025). ***SACA: Selective Attention-Based Clustering Algorithm***. Under review at Knowledge-Based Systems journal. [**Paper**]
- Vafadarnikjoo, A., **Ghaedi, R.**, Imran, N. (2025). ***A Text Mining and Topic Modelling Analysis of UK Modern Slavery Act Statements to Assess Effectiveness of Grievance Disclosures***. Under review at the *International Journal of Computational Intelligence Systems*.
- **Ghaedi, R.**, Farhang, D., Vafadarnikjoo, A. (2025). *Tentative title:* ***A Multi-Agent AI Framework for Forced Labour Risk Assessment in Global Supply Chains***. Manuscript in preparation (experiments completed; writing in progress). Target journal: *International Journal of Production Research* (Special Issue: The Agentic Supply Chain: Entering a New Era in AI-Driven Supply Chain Management).
- Vafadarnikjoo, A., **Ghaedi, R.** (2025). ***A Machine Learning and Simulation Approach for Improving Supply Chain Resilience***. Book chapter under review. Target book: *Computational Intelligence Techniques for Sustainable Supply Chain Management*. [**GitHub**]

Research and Professional Experience

Research Associate **2024 – Present**
University of Sheffield *Sheffield, UK*

- Developed a **multi-agent AI system** for automated corporate ethics benchmarking, enabling rapid assessment of forced labour risks across global ICT supply chains through parallel data processing and predictive analytics.
- Applied NLP and topic modeling **(LDA, BERTopic)** to analyze UK Modern Slavery Act statements, evaluating grievance mechanism effectiveness through systematic text mining of corporate disclosures.
- Designed and implemented a **Decision Support System** using Best-Worst Method (BWM) and Spanning Tree Enumeration (STE) for complex decision-making in supply chain management contexts.
- Contributing to **multiple interdisciplinary publications** bridging AI, operations management, and social impact research (manuscripts under review and in preparation).

Software Engineer **2014 – 2021**
Novin Asan Tech Mana Company *Shiraz, Iran*

- Developed and maintained Java-based enterprise applications with emphasis on backend logic and relational database integration
- Led the migration of legacy systems to modern frameworks, improving system maintainability and user experience
- Collaborated with cross-functional teams to troubleshoot, deploy, and document software modules

Selected Research Projects

Multimodal Personalized Depression Detection *[GitHub]*
Built a multimodal, personalized depression detection system that fuses voice, facial cues, and personal attributes to predict depression level using attention-based deep learning (MPDD Challenge @ ACM Multimedia 2025).

Corporate Ethics Automation System (Kaggle Competition) *[GitHub]*, *[Kaggle]*
Kaggle Agents Intensive hackathon entry: built a Google ADK automation tool to replicate KnowTheChain benchmark workflows in <60 seconds using PDF extraction, sentiment analysis, correlation analysis, and regression.

RepoForge AI: Agentic Test Generation for Public Python Repositories (Kaggle Project)
Built a multi-agent system (OrchestratorAgent plus specialised agents for ingestion, code analysis, API discovery, retrieval, test planning, and test writing) that scans public Git repositories and generates validated pytest suites. Combined RAG over ChromaDB-indexed code chunks with Gemini-based generation, plus safety checks (secret redaction, unsafe-pattern blocking) to produce reviewable, downloadable test artifacts.

Graph-Attention Network for Cross-Domain FER *[GitHub]*
Integrated ResNet-50 and Graph Attention Networks with adversarial domain alignment (MMD, CORAL, GRL). Achieved 74% mean accuracy across six datasets (RAF-DB, FER2013, CK+, JAFFE, ExpW, SFEW2.0).

LDA-Based Grievance Mechanism Analysis *[GitHub]*
Applied advanced LDA topic modeling to 2,957 grievance documents from UK Modern Slavery Act statements, identifying 18 optimal topics to analyze patterns in corporate disclosure practices.

Technical Skills

Programming & Scripting: Python (Advanced), Java (Advanced), C# (Intermediate)
Machine Learning & AI: Deep Learning, Graph Neural Networks (GNNs), Transfer Learning, Domain Adaptation, Multimodal Learning, Representation Learning
Data Analysis & Visualisation: Advanced Excel, JMP, Power BI, Tableau, SQL, Statistical Analysis, Exploratory Data Analysis (EDA)
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers
Databases & Cloud: PostgreSQL, MySQL, MongoDB, AWS (EC2, S3)
Tools & Development: Git, Jupyter, LaTeX, REST APIs

Professional Affiliations

Peer Reviewer: Discover Artificial Intelligence (Springer Nature)
Peer Reviewer: IEEE Transactions on Consumer Electronics
Student Member: IEEE Computer Society Member (No. 100476059)

Achievements & Certifications

GRE: 328 (Verbal 158, Quantitative 170, AWA 4.0)

The Art and Science of Uncertainty and the Future (Interdisciplinary Summer School) **Jul 2026**